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Crisis and entrepreneurship in Greece: Present, past and evolving trends

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Abstract: The current global socioeconomic crisis and restructuring brings various transformations to the dynamics of entrepreneurship. The competitive capabilities of survival of the different socioeconomic systems in globalization also determine the overall global development prospects. Within the context of the particular Greek structural crisis, this paper attempts to highlight how the entrepreneurship crisis is structurally evolving within the Greek socioeconomic system. In this perspective, this article explores the evolution of the Greek business environment, by highlighting the structural morphology of the Greek entrepreneurial ecosystem and noting the required adaptation for the Greek enterprises in "Stra.Tech.Man" terms (synthesis of strategy-technology-management).

Keywords: Global socioeconomic crisis; Entrepreneurship crisis; Greek socioeconomic crisis; Stra.Tech.Man approach

1. Introduction: Crisis, global restructuring and entrepreneurship

Nowadays, it has become increasingly evident that in order to comprehend the current crisis and restructuring of globalization (Aglietta, 2010; Corm, 2013; Graz, 2013; Sapir, 2011; Servet, 2010), in a balanced and valid way, it is imperative to overcome a static or recursive perception of reality. Hence, every fragmentary approach of the international economic dimensions of globalization cannot be considered adequate. On the contrary, it seems that the study of the phenomenon of crisis in globalization needs to constitute a complex and interdisciplinary field of research (Augsburg, 2010; Jacobs & Frickel, 2009).

And this happens because globalization was and remains an inexhaustible dialectic socioeconomic phenomenon, a socioeconomic reality under constant evolutionary transformation, which inevitably reproduces within its core the limits and the perspectives of all the entities which shape and reshape it incessantly (Adda, 2006; Chavance, 2012; Dulong, 2012; Lahire, 2014; Moreau Defarges, 2003; Norrie, 2009).

The dynamics of crisis is still a scientific discipline that remains ambiguous and controversial (Amable, 2017; Vloudos et al., 2018). However, the current global crisis brings to the foreground

the necessary structural transformation of the global socioeconomic reality while imposing an organic and irreversible procedure that drastically changes the global system (Robert & Yoguel, 2016; Scazzieri, 2018).

Under these circumstances, the meaning of entrepreneurship per se evolves into a new, transformed context (Endres & Woods, 2010; Mack & Mayer, 2016). The capitalist firm is a "living" socioeconomic organism, a dialectic socioeconomic entity, which forms and reforms systematically the environment in which it gets born and survives (Iansiti & Levien, 2004; Moore, 1997). And this evolutionary perception of the capitalist firm breaks down every relation to the conventional mechanistic neoclassical interpretation, namely, a consideration that assumes that the capitalist firm is a simple static transformational mechanism with quantitatively defined inputs and outputs (Cahuc, 1998; Favereau, 1989; Gabrié & Jacquier, 1994; Hodgson, 1998; Holmstrom, 1999; Holmstrom & Roberts, 1998).

On the contrary, this article perceives the firm as an active subject and co-creator of the socioeconomic systems that host its action, in a global inter-sectorial level (Mann, 2011; Saviotti & Pyka, 2004). Thus, a realistic interpretation of firm dynamics and transformation is critical for the study of the level of development/underdevelopment of any socioeconomic system, in any spatial level of analysis.

In this context, the concept of competitiveness and its claim seems to become an issue of particular importance (Bhawsar & Chattopadhyay, 2015; Peneder, 2016). It also becomes evident that the systematic interpretation of competitiveness incorporates and synthesizes both the firm dynamics and the socioeconomic fields that host it, along with the inter-sectorial structures formed in a global level (Crozet & Lafourcade, 2009; Gilli, Mazzanti, & Nicolli, 2013; Hidalgo & Hausmann, 2009).

Therefore, this study conducts a critical review of the respective literature by focusing on how the Greek socioeconomic system, and particularly the entrepreneurship in Greece, is confronted with its systemic competitive deficiencies and its particular opportunities inside the dynamics of the emerging "new globalization." The main question is summarized as follows: How has the entrepreneurship evolved in Greece during the last decades?

The answer given in the current work is expected to form the basis for future study on the development/underdevelopment path and the prospect of the Greek business environment. Towards reaching this conclusion, the methodology of this study goes through the following steps, conducting a critical review of the respective international literature:

- (a) It investigates the entrepreneurship crisis, focusing on the case of Greece;
- (b) It analyzes the evolution of the business environment in Greece;
- (c) It presents the structural morphology of the Greek entrepreneurial ecosystem in Stra.Tech.Man terms; and
- (d) Introduces the necessary conclusions and the specified directions of future research.

2. Entrepreneurship crisis in Greece

A vast number of modern studies have researched the crisis topic concerning entrepreneurship. Some of them seem to draw useful conclusions:

- The study of Peris-Ortiz et al. (2014, p. 1), which relates entrepreneurs with their experience, specific business management skills and knowledge, their innovation practices, attitude, and perception of opportunities, postulates that entrepreneurs with these characteristics and practices, embodying entrepreneurship in the fullest sense, will maintain an entrepreneurial attitude in situations of economic crisis.
- Giotopoulos et al. (2017b, p. 927) suggest that given the critical role that ambitious entrepreneurs are likely to perform in times of crisis, creating the right conditions for the emergence and support of innovative, high-growth and export-driven new ventures may be a selective and more compelling growth strategy. In this direction, the implementation of structural reforms in product and labor markets, the establishment of entrepreneurial universities, and support of female entrepreneurship should be policy priorities. These actions could also mitigate the effect of fear of failure on ambitious entrepreneurship as they may reduce the risk of starting a new business. Such mechanisms and policy tools may help in nurturing and leveraging those crucial quality entrepreneurial elements that will foster value creation of new ventures and increase their multiplying effect on economic growth in Europe.
- Korsgaard et al. (2016, p. 178) emphasize the spatial aspect and argue that their analysis indicates that the opportunistic discovery view and, to some extent, the resourcefulness view are both inadequate as conceptual platforms for entrepreneurial responses to the economic, environmental and socio-spatial crisis. Instead, they develop the "entrepreneurship as re-sourcing" as an alternative perspective on entrepreneurship. This perspective emphasizes the importance of building regional-level resilience through the entrepreneurial activity that sources resources from new places and uses these resources to create multiple forms of value.

More specifically, for the entrepreneurship crisis in Greece in recent years, the following approaches are useful:

- I. Giotopoulos et al. (2017a, p. 540) highlight the role of the human capital of entrepreneurs for ambitious new ventures, even though different aspects of entrepreneurs' human capital appear to matter in crisis and non-crisis periods. While work experience seems to positively affect the probability of being an ambitious entrepreneur during the crisis years, educational attainment of entrepreneurs appears to be an essential driver of growth-oriented entrepreneurship only in the pre-crisis period. This result, though somewhat strange, may be explained by the accelerated during the crisis period in Greece "brain-drain" phenomenon and the work options in established firms that may be available for well-educated and highly skilled people even in adverse times.
- II. Vassiliadis and Vassiliadis (2014, p. 246), by searching the aspect of family-business entrepreneurship suggest that family firms make an essential contribution to the national economy and bring long-term stability. A critical advantage of a family business is its ability to survive in times of economic crisis. Greece has one of the most significant numbers of SMEs within the EU, and most of them are family businesses. The most characteristic feature in Greece is that the members of the family own, manage, and influence the business. As far as corporate governance is concerned, Greek family-owned businesses are under-governed. The primary obstacles faced by the Greek family businesses are human resources issues, bureaucracy, unstable tax environment, as well as family conflicts and lack of succession planning. Preparing for

a business transfer to the next generation is a long and complicated process and can entail several obstacles. Professional support bodies in Greece would help the transfer of family businesses to the next generation considerably.

- III. Herrmann and Kritikos (2013, p. 20) explore how Greece can move towards an innovation-based economy, arguing that Greece's Euro-zone membership may have given a false impression that it is an innovation-driven economy. The Greek economy is not, however, since it faces not only institutional but also severe structural deficits with a small industrial basis, low export ratio, small businesses, and many closed professions. If decreasing labor costs and further institutional reforms were to be the only active policy, then Greece's future would be a low wage economy with an extended workbench of other innovative economies. Greece can only become prosperous if it also uses its comparative advantages beyond tourism, trade, and agriculture.
- IV. Kaplanoglou et al. (2016, p. 432) present the results of a survey of 550 small and micro enterprises in Greece, a country facing one of the most significant tax gaps in the developed world, regarding their tax compliance behavior, and suggesting that trust seems to play the most significant role in increasing intended compliance with tax obligations and in deterring strategic tax evasion. The effect of powerful tax authorities on tax behavior depends on the level of trust to government. Power of tax authorities increases voluntary compliance only in high-trust settings. It seems, therefore, that people are more willing to entrust their money to a government that uses it wisely if an effective tax administration supports the latter.
- V. Giannacourou et al. (2015, p. 548) use data from case studies of Greek SMEs and explore the impact of the crisis on managers' perceptions of uncertainty by comparing their pre-crisis and after crisis perceptions. In particular, before the outbreak of the crisis, most managers reported that levels of uncertainty were medium-to-low in all measured parameters, although the answers of the managers of smaller firms centered on the middle of the scale. After the outbreak of the crisis, most managers reported an increase in uncertainty in all categories comprising environmental uncertainty, although the increase reported by managers of medium-sized firms was slightly lower.
- VI. Sainis et al. (2016, p. 327) explore the willingness of ISO certified Greek SMEs to continue developing their total quality management under crisis conditions. By using as quality criteria for measuring the TQM implementation level the "organizational performance assessment," the "organizational culture," the quality processes and the quality tools and techniques, conclude that the Greek SMEs continue their quality journey emphasizing cultural dimensions. The low rates recognized for quality tools and processes implemented, show that ISO certified SMEs reinforce TQM implementation but not in such a level as to contribute to a "continuous improvement" process.
- VII. Williams and Vorley (2015, p. 28), through the case study of Greece, demonstrate how the institutional environment has changed in light of the crisis and the resultant response of entrepreneurs to these changes. They draw in-depth interviews with entrepreneurs and find that changes to institutions have served to limit entrepreneurial activity rather than enhance it and that this has worsened amid the crisis. They argue that this phenomenon will detrimentally impact Greece's ability to navigate out of the crisis and regain competitiveness in the longer term.

- VIII. Giotopoulos and Vettas (2018, p. 877) explore the antecedents of export-oriented entrepreneurship, such as the demographic characteristics, perceptions, and dynamism of entrepreneurs, in Greece at times of crisis, using a micro-level data analysis. They suggest that in the light of the recent economic crisis and the next recessionary economic cycle that affected the Greek economy, the need for restructuring the productive entrepreneurial system towards a dynamic and export-oriented critical mass is more eminent than ever in order to create value, generate jobs, and facilitate economic recovery.
- IX. Kapitsinis (2019) employs a comparative analysis of the pre- and post-crisis movements of Greek small and medium-sized enterprises (SMEs) to Bulgaria in order to examine the impact of the crisis and the applied public policy on firm-internal relocation factors, such as size, sector and relocation incentive, and the effects of relocation on business performance. Greek SME movements to Bulgaria have recently increased considerably due to the adverse effects of the crisis on the Greek economy. Results demonstrate that, while in the pre-crisis period many Greek businesspeople viewed relocation to Bulgaria as an entrepreneurial opportunity for expansion since 2007 relocation is a necessity for the vast majority of Greek entrepreneurs in order to stay in business.

3. The entrepreneurship environment in Greece

At the same time, in order to comprehend the unique environment of entrepreneurship in Greece nowadays, it is useful to consider the approach of Global Entrepreneurship Monitor (GEM), which is being published since 1999 and provides the respective publication for Greece since 2003. More specifically, the most crucial index and the innovation of this approach is the "early stage entrepreneurship" which calculates the percentage of the 18-64 population who are either a nascent entrepreneur or owner-manager of a new business (Bosma & Kelley, 2018, p. 138).

More specifically, the latest report (Bosma & Kelley, 2018, p. 80), which profiles 49 economies with respect to demographics, their potential impact, the diversity of forms they take, and their longer-term sustainability, reaches the conclusion for Greece that the improvement in early-stage entrepreneurship during 2018 is closely related to the rebound of the Greek economy after a period of profound and prolonged economic crisis. The growth rate in 2018 is forecasted to reach 2%, (1.4% in 2017). The most important milestone of 2018 was the fact that in August, an eight-year cycle of three successive programs of economic support and adjustment programs completed. The economy achieved a significant amount of adjustment and rebalancing, increased its exports and investments, although domestic financing conditions remain very weak. The report states that the economy requires policies aimed at encouraging people with high educational levels to promote entrepreneurship. These policies should focus not just on a mere quantitative increase of new ventures, but on the quality of these ventures, i.e., their innovation and growth potential.

The GEM report classifies the countries in four regional groups (East and South Asia, Europe and North America, Latin America and Caribbean, Middle East and Africa), and further differentiates them in terms of their income level (low-income, middle-income, high-income); Greece falls into the high-income category. In the following table taken from the last GEM report, among about 50 economies, Greece is classified in the lowest positions (see Table 1).

Table 1: Entrepreneurial behavior and attitudes in Greece, according to the Global Entrepreneurship Monitor (Bosma & Kelley, 2018, p. 80)

Self-Perceptions About Entrepreneurship		
	Value	Rank
Perceived opportunities	19.2	48/49
Perceived capabilities	46.4	31/49
Fear of failure	57.8	3/49
Entrepreneurial intentions	7.5	41/48
Activity		
	Value	Rank
Total early-stage Entrepreneurial Activity (TEA)		
TEA 2018	6.4	38T/48
TEA 2017	4.8	49/54
TEA 2016	5.7	57T/65
Established business ownership rate	10.8	14/48
Entrepreneurial Employee Activity – EEA	1.8	36/49
Motivational Index		
	Value	Rank/48
Improvement-Driven Opportunity/Necessity Motive	3.0	16
Gender Equality		
	Value	Rank/48
Female/Male TEA Ratio	0.45	44
Female/Male Opportunity Ratio	0.78	44T
Entrepreneurship Impact		
	Value	Rank/48
Job expectations (6+)	9.7	38
Innovation	28.4	18
Industry (% in Business Services Sector)	12.0	27
Societal Value About Entrepreneurship		
	Value	Rank/47
High status to entrepreneurs	67.8	33
Entrepreneurship a good career choice	64.9	20

Concerning the rest indices of Table 1, the GEM calculates them by extracting empirical data. Based on the latest report (Bosma & Kelley, 2018, pp. 138–139), these indices take into account the following dimensions in order to produce the final indicative scores:

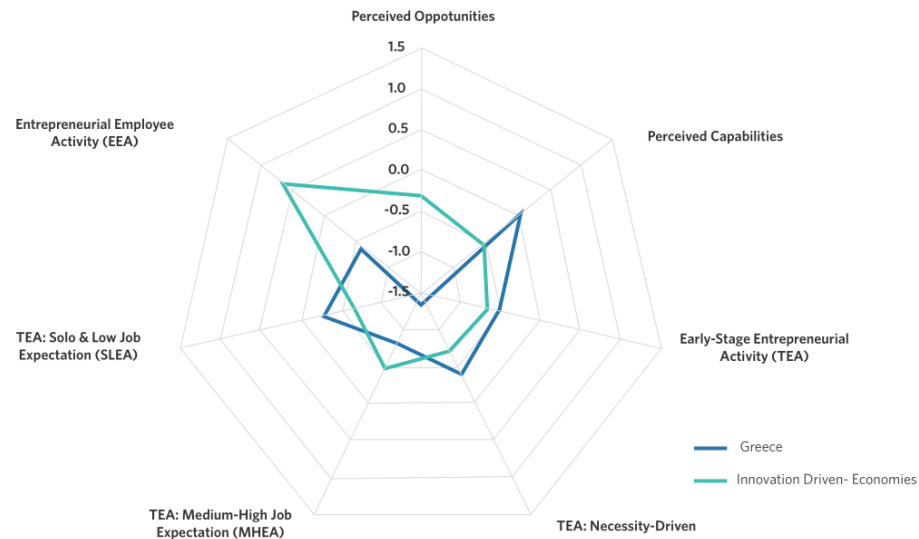
- Perceived opportunities: Percentage of the 18-64 population who see opportunities to start a firm in the area where they live.
- Perceived capabilities: Percentage of the 18-64 population who believe they have the required skills and knowledge to start a business.
- Fear of failure: Percentage of the 18-64 population with perceived opportunities who also indicate that fear of failure would prevent them from setting up a business
- Entrepreneurial intentions: Percentage of the 18-64 population (individuals involved in any stage of entrepreneurial activity excluded) who intend to start a business within three years.
- Established business ownership rate: indicates the percentage of the 18-64 population who are currently owner-manager of an established business, i.e., owning and managing a running business that has paid salaries, wages, or any other payments to the owners for more than 42 months.
- Entrepreneurial Employee Activity – EEA: percentage of the 18-64 population who, as employees, have been involved in entrepreneurial activities such as developing or launching new goods or services, or setting up a new business unit, a new establishment, or a subsidiary.

- Improvement-Driven Opportunity/Necessity Motive: percentage of those involved in TEA who (i) state they are driven by opportunity as opposed to having no better options for work; and (ii) who indicate the main driver for being involved in this opportunity is being independent or increasing their income, rather than just maintaining their income.
- High status to entrepreneurs: Percentage of the 18-64 population who agree with the statement that in their country, successful entrepreneurs receive high status.
- Entrepreneurship a good career choice: Percentage of the 18-64 population who agree with the statement that in their country, most people consider starting a business as a desirable career choice.

Furthermore, some older GEM studies, which comment on the general business environment in Greece, present some useful conclusions.

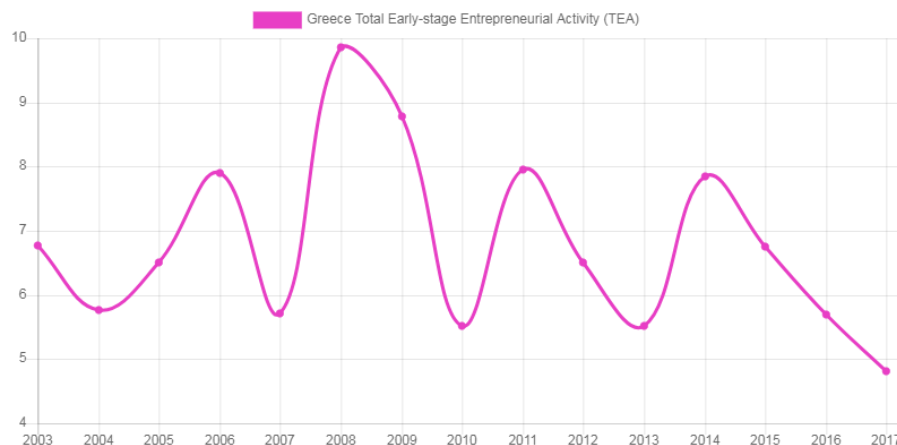
- According to the 2010 report (Kelley, Bosma, & Amorós, 2010), the established business rate in Greece is significantly high. They explain that by stating that, to some extent, agriculture dominates the economy in Greece.
- According to the 2011 report (Bosma, Wennekers, & Amorós, 2011), there is an absence of reforms in Greece, whose economy lacks dynamism. Except for physical and commercial infrastructure, as well as support for entrepreneurship, all other elements constituting the entrepreneurship framework conditions are unfavorable, since no significant restructuring or reform has taken place over the past few years. Also, the report notices that Greece has a very high level of established business owners and to have so many people involved as business owners in comparison to not just other innovation-driven economies but also to Greece's own early-stage entrepreneurial activity points at a limited degree of dynamism.
- As per the 2013 report (Amorós and Bosma, 2013, p. 53), business opportunities in Greece are dramatically low because of the crisis, whereas the business skills are very high. The report states that the entrepreneurial profile of Greece differs quite a lot from the average profile of an innovation-driven economy. Even though Greece's TEA rate is slightly higher than (but still comparable to) other economies, other indicators show that due to the crisis in Greece, perceived opportunities to start a business are dramatically low, even though perceived capabilities are quite high. Also, the nature of entrepreneurial activities tends to be one of low ambition and relatively driven by necessity. Entrepreneurial employee activity, as measured by the GEM 2011 assessment, is quite low in Greece (see Figure 1).

Figure 1: The entrepreneurial profile of Greece (Amorós & Bosma, 2013, p. 54)



Also, the following diagram for the early stage entrepreneurship in Greece from 2003 to 2017, illustrates substantial fluctuations, which culminated before the outbreak of world crisis in 2008, whereas today this specific index has reached the lowest value ever (see Figure 2).

Figure 2: Total early-stage Entrepreneurial Activity in Greece (Global Entrepreneurship Monitor, 2019)



According to Tsakanikas et al. (2018, p. 5-7), the authors of the 2017 GEM research focused on the case of Greece, the characteristics of early-stage entrepreneurship in 2017 are the following. In 2017, the percentage of the population aged 18-64 at the early stages of entrepreneurial activity (including self-employment) dropped to 4.8% (approximately 320 thousand people) from 5.7% (approximately 380 thousand) in 2016. Some 56% of early-stage entrepreneurs are new entrepreneurs, while the remaining are nascent entrepreneurs. About 17% of the population aged 18-64 (1.13 million people) has some relation with entrepreneurship, either in early or in subsequent stages. The rate of the population that shut down an entrepreneurial venture in 2017 reached 4.2% (approximately 310 thousand people), higher than the corresponding rate in 2016 (3.8%) and much higher than the average of innovation-driven economies (2.2%). 29% of early-stage entrepreneurs (about 90 thousand people) are necessity-driven entrepreneurs, while 37% (about 120.000 thousand people) are opportunity-driven entrepreneurs. In terms of age, in 2017, the higher participation in early stages of entrepreneurship was from the age group 35-44 years old, (7.6% versus 11.3% in

innovation-driven economies). The percentage of female early-stage entrepreneurship dropped at 3.9% (about 131 thousand women) from 4.8% in 2016, while the percentage for men dropped to 5.7% (about 185 thousand men) from 6.6% in 2016. In 2017, 41.7% of early-stage entrepreneurs had at least a degree of tertiary education, presenting a higher share compared to 2016, while another 11.5% had a postgraduate degree. Finally, 3.2% of people aged 18-64 (approximately 214 thousand people) declared that they were informal investors in another person's business venture, a rate slightly lower compared to 2016, but also lower than the average of innovation-driven economies (5.4%).

4. The structural morphology of the entrepreneurial ecosystem in Greece

The enterprises that operate in Greece, like living/adaptive organisms, are not all of the same kind. All enterprises in Greece as living organisms, as “animals” of different species, do not share the same evolutionary “physiology” (Vlados, 2004, 2005, 2012). The ‘fauna’ of these enterprises, as well as their socioeconomic ecosystem, is not at all the same; and as it seems, nor will it easily catch up in the future with that of the developed capitalist countries.

The specific dimensions of the business environment in Greece keep changing in an incessant evolutionary way. All the socioeconomic “islets” on the planet, namely, the partial socioeconomic systems, are coming even closer because of the dynamics of globalization. The socioeconomic systems come closer, without having their structural socioeconomic “idiomorphism” absorbed or wiped out. On the contrary, their structural singularity keeps reproducing evolutionarily in the restructuring of globalization (Ladhari, Souiden, & Choi, 2015; Roudometof, 2014; Scherer, Palazzo, & Seidl, 2013).

A part of contemporary literature and research, however, falls into an “interpretive trap.” A field of modern economic science, either neoclassical or neo-Marxist, does not seem to understand what exactly globalization is per se. Thus, it cannot gain a thorough insight into the structural and “physiological” evolution of firms in this context. On the contrary, the interpretation that conceives firms as living organisms, in a biological-type theoretical perspective can be more thorough analytically (Geus, 2002; Meyer & Davis, 2003; Zeleny, 2010).

4.1. How do different types of enterprises in Greece think and operate?

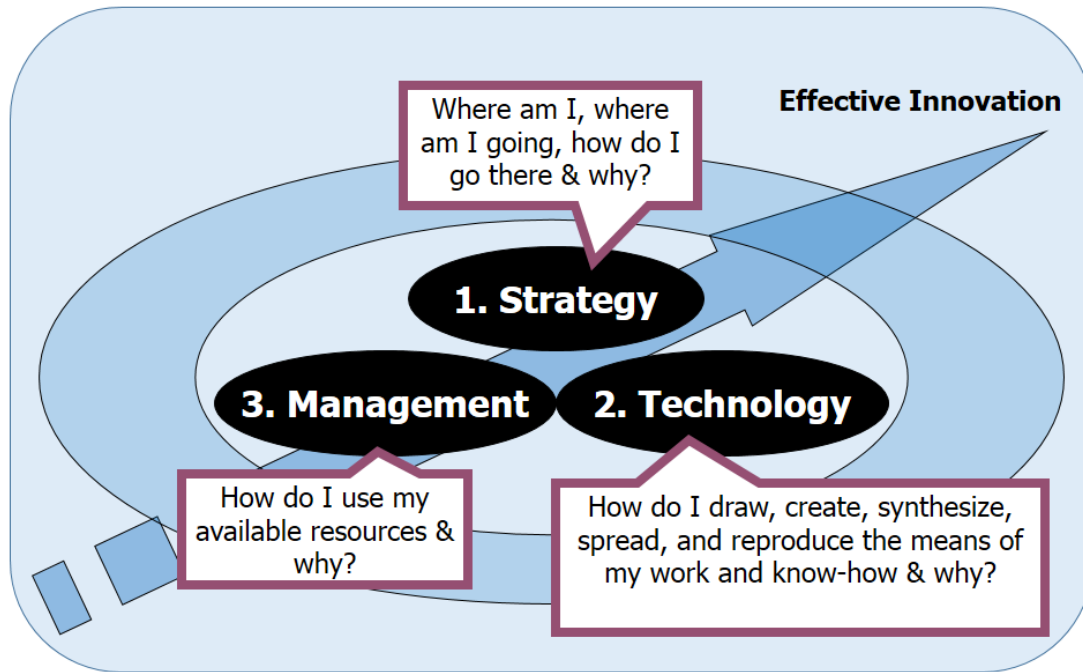
Based on the findings of previous studies (Βλάχος, 2006, 2016), which continue to get validated on the field, the enterprises that operate in Greece in different sectors of economic activity, form a dynamic three-pole: namely, the entrepreneurial ecosystem in Greece gets evolutionarily polarized among three central and distinct “Stra.Tech.Man physiologies.”

In particular, according to Vlados (2004, 2005), a socioeconomic organization is perceived as an evolutionary entity that synthesizes internally three co-evolving spheres: its strategy (Stra), technology (Tech), and management (Man). A threefold set of profound questions—always and necessarily—defines and advances the organization's specific “Stra.Tech.Man” dynamic potential of innovation:

- i. Strategy corresponds to the set of questions, “where am I, where am I going, how do I go there and why?”
- ii. Technology corresponds to the set of questions “how do I draw, create, synthesize, spread, and reproduce the means of my work and know-how and why?”

- iii. Management corresponds to the set of questions “how do I use my available resources and why?” (see Figure 3)

Figure 3: The evolutionary Stra.Tech.Man core of the enterprise (Βλάχος, 2006)



Each entrepreneurial type (Siakas et al., 2014) perceives business reality (business philosophy) and operates (business procedures) in a different way, both in terms of management, technology, and strategy. These differences always emerge in every link of their operational chain, as illustrated by the Tables that follow (see Tables 2, 3, and 4).

Table 2: The central types of management in today’s Greek enterprises (Βλάχος, 2006)

Type of Management	“Monadocentric” Enterprise	“Massive” Enterprise	“Flexible” Enterprise
CENTRAL LOGIC	PRACTICAL EXPERIENCE	SPECIALIZATION	PARTICIPATION
Planning	Oriented to immediate response, providing short term solutions.	It sets long-term quantitative goals and norms.	It follows a constant spirit of quality and continuity: systemic unification of short-term and long-term goals.
Organization	Based on the temporariness of functional roles and overlapping of responsibilities.	Based on constant, precise, and generally inflexible hierarchy/organization chart.	Based on the autonomy of functional teams, with constant efforts of decentralization.
Staffing	It focuses on the search of a worker that has practical experience of the field.	It focuses on the search of qualified specialist and worker with minimum demands.	It focuses on the search for a creative worker/collaborator.
Development of human resources	Perceived as non-relevant; the enterprise does not undertake any systematic action.	It focuses on specific, narrow, and segmented operational areas.	It extends on top of the continuous cognitive enrichment of all levels of hierarchy and across all departments.

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Motivation	It focuses on the reproduction of “safe relations” through the activation of an unusual character of a “traditional paternalistic family.”	It is based on simple and of limited extent, economic incentives.	It focuses on idealistic incentives with a long-term perspective.
Leadership	It is based on the personality of the “boss.”	It is based on strict and detailed rules and routines.	It is based on the support of open-mindedness, with a character that encourages advice-giving instead of order-giving.
Control	Through a fluctuating strictness and personalized regulatory interventions.	It is centralized, based mainly on clear and formal reports.	It is decentralized and internalized: an effort towards assimilating control to the project and “inside” the worker at the same time.
Communication and coordination	Mainly spontaneous and informal: activated after the creation of the problem.	Mainly of a vertical type, from top to bottom: the flow of information happens in a linear and fragmentary way.	Mainly of a mixed type: vertical, horizontal, and diagonal in the organization chart. Diffusion of information in interactive networks.

Table 3: The central types of technology in today’s Greek enterprises (Βλάχος, 2006)

Type of Technology	“Monadocentric” Enterprise	“Massive” Enterprise	“Flexible” Enterprise
CENTRAL LOGIC	SPORADIC	LINEAR	NETWORK-BASED
Technological vigilance	It focuses on the instantaneous exploitation of technological opportunities (usually total lack of systematic focus on the evolving technological environment).	It focuses on the systematic exploitation of pre-installed technological frameworks (focus mainly on the internal technological tradition of the enterprise).	It focuses on the systemic and interactive exploitation of the technological data (effort towards progressive unification of the internal and external technological environment of the enterprise).
Technology acquisition	It is based on a static selectivity: “the trustworthy supplier of technology chooses for the company.”	It is based on a one-way selectivity through “small leaps.”	It is based on a multifarious selectivity with a dynamic and constant character.
Internal creation of technology	Very limited: it focuses on the search for “better standardization.”	Relatively limited: it focuses on the continuous increase of quantitative performance.	Well-developed and durable: focused on the continuous improvement of all the qualitative dimensions of the enterprise.
Technology assimilation	It is perceived as a “point-to-point” and enclosed logic of the working positions; works for immediate processing.	It is perceived according to a mechanistic logic, which usually is limited by narrow functional/departmental terms.	It is perceived according to an organic logic: an effort towards parallel integration in terms of worker—team—the entire organization.

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Technology diffusion	Fragmented and enclosed in specific individual tasks: limited to the craftsman’s “knowledge.”	It is enclosed in specific productive functions/departments.	Cross-departmental, holistic, and with multiple bi-directional orientations.
Technology application	Dominated by a focus on personalized work: the “experienced craftsman” is the principal agent.	Dominated by a focus on the formal administrative structure: the highest in the rank executive is the principal agent.	Dominated by a focus on cross-departmental exploitation: “the creative collaborator of the nearby department” is the principal agent.
Assessment of technological effort	It focuses on sporadic evaluations: point to point, task to task, worker to worker, but rarely as a whole.	It focuses on evaluations of technical efficiency: usually strictly perceived as a race for quantitative improvements.	It focuses on the evaluation of total technological efficiency: perceived in a multi-synthetic way and with a cohesive qualitative perspective.
Technological synthesis	It is based on the individual skills and talent of the technician.	It is based on procedures and on “offices of methods” pre-installed and typically responsible.	It is based on the collaborative creation and the “collective talent” of all the action teams.

Table 4: The central types of strategy in today’s Greek enterprises (Βλάχος, 2006)

Type of Strategy	“Monadocentric” Enterprise	“Massive” Enterprise	“Flexible” Enterprise
CENTRAL LOGIC	INTUITIVE	MECHANISTIC	EVOLUTIONARY
Total strategic perspective	The pursuit of short-term market success, avoiding immediate risks.	A procedure of repetitive quantitative plans is guiding the enterprise.	It progressively assimilates a unifying spirit for the totality of the dynamic dimensions surrounding the enterprise.
Nature of business plan	It is based on an implicit and informal procedure that relies on the “hidden thoughts” of the business owner.	It is based on a systematic and mainly bureaucratized methodology.	It is based on an open and multi-collective methodology, to synthesize simultaneously precise and “underground” data.
Monitoring of the general external business environment	It monitors with an occasional/superficial character.	It monitors with an inflexible “calendar type” character.	It monitors with a constant and strongly reflective character in systemic terms.
Monitoring of the direct, sectoral and local competitive environment of the enterprise	It is realized, usually, in terms of an emergency action task.	It is realized, usually, in terms of directly contradictory comprehension of competitiveness (as a zero-sum game).	It is realized, usually, in terms of progressive “endogenization” of changes (as a positive-sum game).
Monitoring of the internal business environment	It is concentrated on the direct initiative of the owner/boss without a clear and explicit method.	It is concentrated on the evaluation of the functional efficiency, based on detailed administrative audits.	It is concentrated on the evaluation of the total efficiency of the organization, based on integrated quantitative/qualitative audits.
The exploitation of partial strategic dimensions	It focuses on the dimension of immediate market profitability with repetitive character and nothing more.	It focuses on the simple combination of internal functional advantages with the needs of the	It focuses on the cross-fertilization of the inherent advantages of the enterprise with the evolution of the broader trends of the market

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		market (a comparatively static logic).	and the economy. The constant pursuit of “hidden synergies.”
Assessment of strategic analysis and synthesis	Ambiguous and person-oriented: relying on the “individuality” of the owner.	Standardized and mainly dichotomous: relying on the reproduction of the interests of the top executives and secondarily on the shareholders.	Multilevel and functionally integrated into the entire enterprise: established on “strategic democracy” and in the quest for continuous entrepreneurial evolution.
The contribution of strategy in the search for entrepreneurial success and excellence	The strategy is an unusual mixed perspective: opportunistic and conservative at the same time. Capital exploitation remains, in most cases, short-term and “modest.”	The strategy is an often introverted apprehension. Capital exploitation remains partitioned between short-term and long-term.	The strategy is a dialectical (co-evolutionary) apprehension of the internal and external environment. Capital exploitation becomes unified between the short- and the long-term, without one-dimensional enforcement of immediate maximization of profitability.

The tables above that describe the central types of strategy, technology, and management in today’s Greek enterprises are useful to understand some critical dimensions:

- First of all, they unravel the fact that the differentiation of the three different firm types is neither superficial and coincidental nor easily reversible. On the contrary, their differentiation is structural and substantially physiological.
- They also reveal the deeper structural cohesion of these three poles. These three types of enterprises continue to operate as structural poles of the entrepreneurial ecosystem in Greece. Even though the three types do not express a static classification, they do manage to clarify the reproduction of the business ecosystem in Greece cohesively.
- The mixed physiological Stra.Tech.Man types (hybrids) are both existent and possible; however, they always come under an affinity limitation in the combination of different Stra.Tech.Man logic that they synthesize (Vlados & Chatzinikolaou, Forthcoming; Vlados & Chatzinikolaou, 2019).

All the “living” firms in Greece cannot incorporate big “selective openings,” and they cannot cope with significant internal physiological distances in the individual Stra.Tech.Man terms. It is also not possible for these enterprises to make big physiological leaps, from one day to the next. The sustainable evolution of a firm is always carried out under the condition of co-evolution of these three Stra.Tech.Man spheres that characterizes the physiological root. Moreover, in this way, the automatic shifts and transformational “miracles” of any organization do not seem possible (Marshall, 1890).

4.2. How do the enterprises survive in Greece?

Under the current circumstances, the “monadocentric” family type firms seem to keep comprising the backbone of the productive system in Greece. In particular, they tend to give answers to the links of the chain of their thought and action in a “reflective way.” These answers are least systematic and consistent with a cohesive entrepreneurial perspective. And not only that, but their “business logic” is difficult to reconcile with any stereotypical image of the rational capitalistic efficiency.

As it happens with all the firms, the “monadocentric” type ones survive in the globalized competitiveness according to the composite-triple Stra.Tech.Man dynamics of competitiveness. Namely, they combine their physiology (what exactly they are) with the properties of the “idiomorphous” external socioeconomic environment that hosts them, always operating and “fitting” in specific globalized sectors.

Why do they continue to constitute the majority in Greece, even in the current crisis conditions? Simply because the current socioeconomic conditions in Greece keeps favoring them:

- With the factors of production market not particularly fertile and knowledge-oriented. In particular, the labor market does not have relatively high skills and, on many occasions, it is contradictory and inflexible. At the same time, the labor market becomes gradually “dual” (Christopoulou & Monastiriotes, 2016), not only between the urban and rural population but primarily between the private and the public sector.
- With a product/services market that is neither particularly fertile nor especially demanding in terms of quality. A market where the “about” in terms of quality and price survives domestically, however, it finds great difficulty to be exported and rewarded in foreign markets (Bournakis, 2014; Chrysomallidis & Tsakanikas, 2017; Giotopoulos & Vettas, 2018; Klonaris & Agiangatzoglou, 2017; Pitelis, 2012).
- With a knowledge and innovation environment generally poor, weak, and enclosed, continually reproducing its various “pathogenies.” At the same time, the work mentality is generally “reassured,” while entrepreneurship has significant deficiencies in entrepreneurial knowledge and skills (Cohen, Naoum, & Vlismas, 2014; Panagiotakopoulos, 2015).
- With a rich cultural background that, however, is “ankylosed” by a problematic mentality of political irresponsibility, which reproduces entrenched interests and infertile introversion.
- With a natural/environmental background (Koutalakis, 2011; Lekakis & Kousis, 2013) poorly managed and increasingly threatened. Also, with a demographic perspective also gradually worsened (Ifanti et al., 2014; Labrianidis & Vogiatzis, 2013).
- Finally, with a state-intervention logic most of the time intentionally labyrinthical, “dyspraxic” and ineffective, despite the passage through the painful adjustment period of the memorandums.

4.3. How do the enterprises in Greece seem to adapt?

So the fact that the “monadocentric” enterprises dominate in numbers in Greece should not come as any surprise. They constitute the majority of enterprises because they manage to “fit” with the conditions of the national socioeconomic system, even in the conditions of its structural crisis. However, the critical issue is to what extent and how will they carry on achieving to “fit” (Carmeli, Gelbard, & Gefen, 2010; Prajogo, 2016) in the rapidly globalized future.

Fa long time now, the modern game of competitiveness in the globalization, is not a simple case. It is not enough to “fit.” The enterprise also has to “stretch” in resources and competitive skills. At this point, the major problem for the reproduction of “monadocentric” enterprises—and not exclusively them—arises in today’s Greece. To a large extent, the “monadocentric” enterprises in Greece today, show significant weakness in “stretching” creatively in different

sectors. This stake of “stretching” brings to the foreground the dynamics of globalization of the sectors of economic activity. For this reason, the problem of evolution of the productive backbone of the Greek economy must be taking into account a globalized perspective (Andreoni & Scazzieri, 2014; Lundvall, 2011).

In practice, nowadays, a big part of the “monadocentric” enterprises in Greece —along with a large part of the “massive” enterprises—do not manage to “stretch” upwards; it retreats trying to defend itself.

However, what does this mean? It seems that a large part of firms in Greece, sometimes subconsciously, keep retreating within the globalized hierarchy of their sectors of activity, moving towards more accessible, weaker, less demanding but less profitable subsectors of the global market (Dauth & Suedekum, 2016; Mahmood, Ghani, & ud Din, 2015). Since all sectors of economic activity do not cease to get transformed qualitatively in a global scale, a big part of firms in Greece seem to move to less productive and promising domains of the global “construction site”; towards less fertile, in terms of added value and strategic interest, links of production within the globalized division of labour (Gill, 2016; Poli, 2010). The most vital part is not the sector of activity per se as a field of reference. The critical issue is that the national productive backbone seems to keep shifting to less productive and profitable global sectors, without being able to protect its domestic markets effectively.

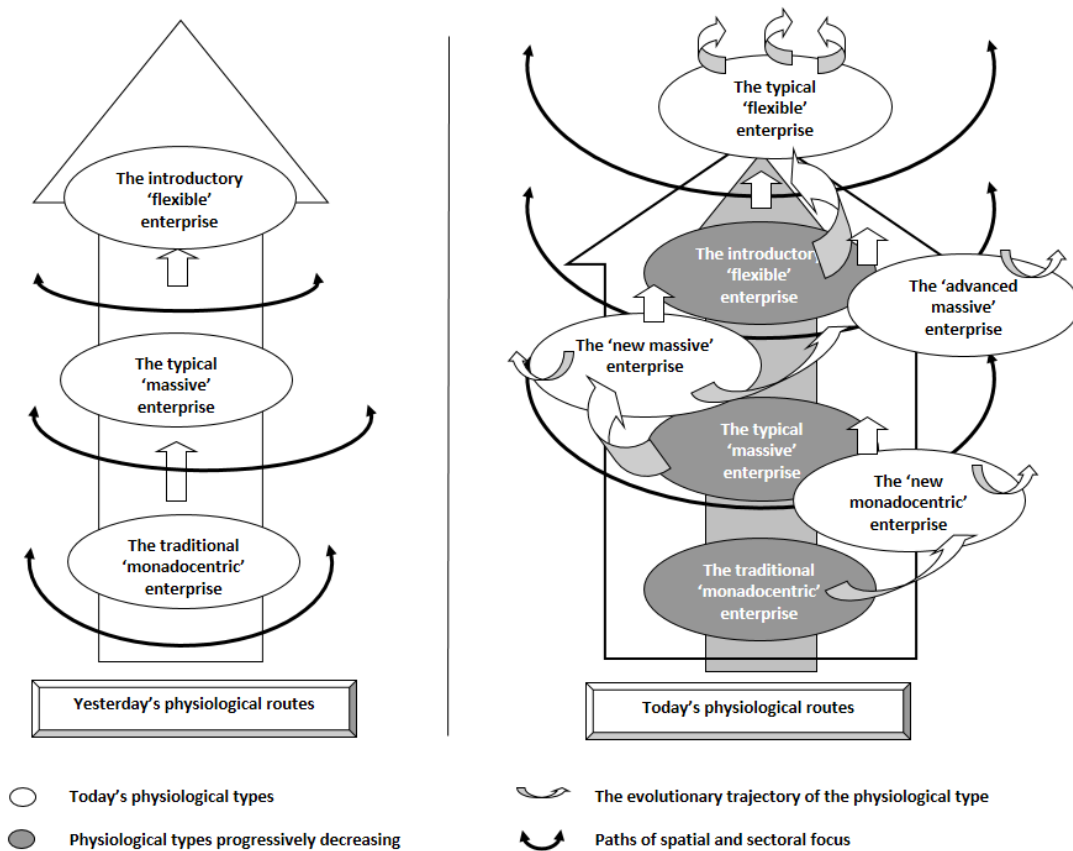
In brief, the entire national productive system seems to keep being oriented for decades—or instead pushed passively—to the less demanding part, in terms of quality and profitability, of the globalized sectors of production. The problem is that the Greek enterprises are led to produce what they can produce as well and those who are significantly “cheaper” than them; and to this point, the critical issue of lack of competitiveness of the productive system seems obvious (Altomonte & Békés, 2016; Annoni et al., 2017).

4.4. How do the enterprises evolve in Greece?

Ultimately, the issue of quality of the Greek enterprises in broad, cohesive and evolutionary terms, seems to continue being the touchstone over the competitiveness of the national socioeconomic system (Committee of the Regions, 2011; Fransman, 2018). More specifically, the claim of better quality is mostly a matter of the Greek firms’ entire organization and function—a matter of their particular evolutionary Stra.Tech.Man “physiology.” Not only that, but is also a matter of how the surrounding socioeconomic system evolves by targeting the globalized sectorial agglomerations intelligently (Fujita & Thisse, 2013; Nathan & Overman, 2013).

However, there are still some optimistic points. As the heterogeneity of firms increase—in a generally negative cross-sectorial trajectory for the Greek economy—many enterprises manage to improve contrary to expectations. In most cases, these are dynamic organizations—hybrids in Stra.Tech.Man terms. They are mainly new generation business types and models (Bharadwaj et al., 2013; Dahan et al., 2010) (see Figure 4).

Figure 4: The expanding heterogeneity in the pursuit/preservation of competitiveness: the insertion of the Greek enterprises in globalization (Βλάχος, 2006)



The old types/forms of the three “entrepreneurial species” prove to be the most vulnerable in the race of globalized competitiveness; therefore, they are gradually moving away from the competition. The newer, physiologically evolved representatives of the three species prove to be the most resilient and promising. Also, they tend to accept the resort to multinational involvement (Dau, 2013; Monaghan & Tippmann, 2018; Radlo, 2012), alliances and coalitions, which increase the margins of “sectorial refocusing” drastically and sometimes the opportunities of fast physiological evolution.

More specifically, it seems that the enterprises that prove to be more resilient, achieving better performance even in the current economic crisis era, continue to be the following ones:

- The “new monadocentric” enterprise, which is faster and smarter in its choices, with selective assimilation of “massive tools”; it is more extroverted and sometimes relocated to countries that can offer advantages it knows how to use (cheap labor, complicated bureaucracy, such as the Balkans).
- The “new massive” enterprise and the “advanced massive” enterprise—the latter with systematic openings to the flexible model—which manage to exploit progressively, advantages of the Greek national socioeconomic system that have been unexploited so far (common European market, monetary stability, preservation of a relatively developed domestic consumption pattern, specialized scientific workforce which is not expensive any more).

- The authentic and typical "flexible" enterprise, that is, the offspring of the first form of flexible enterprise, which manages to become more aggressive in the claim of high-standard parts of the global market.

For all these hybrid-type "new generation" organizations (Battilana & Dorado, 2010; McMullen, 2018)—and mainly for the "monadocentric" type, which continues to bear the most significant pressures—the goal of the efficient "stretch" in physiological, spatial, and sectoral terms becomes of great importance.

5. Initial conclusions and directions for future research

In order to have a thorough understanding of the crisis of entrepreneurship in Greece, the approach of evolution of the external business environment is not enough. It seems that it is particularly useful an approach in terms of internal environment and analysis of the evolutionary "physiology" of firms.

In terms of the internal and external environment, a double deficiency characterizes the productive backbone in Greece in the last few years:

- On the one hand, the Greek productive system reproduces many structural deficiencies in the external environment, and
- On the other hand, a significant number of Greek firms, particularly the "monadocentric"/family businesses, are unable to find ways of efficient and fast evolution.

This article suggests that this combined and co-evolutionary perspective can pave the way for a new generation of research concerning the evolution of the business ecosystem in Greece, which may offer helpful directions for the configuration of respective structural economic policies.

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